School of Engineering, Arts, Science and Technology

RELATIONAL DATABASES: IMPLEMENTATION

S183038

Word Count: 1084

Table of Contents

Introduction

WC: 99

This report documents the implementation of the relational database designed in part one. The database is responsible for storing and managing the data of a new online video game marketplace, positioned as a competitor to existing platforms such as “Steam”. The report covers the process of converting the logical Entity Relationship Diagram (ERD) into a fully functional MySQL database, alongside the testing and execution of various SQL queries on scripted test data. MySQL Workbench and JetBrains Data Grip were used for database design, query writing, and interacting with the database environment during development and testing (Oracle, 2025; JetBrains, 2025).

Setting up an instance of MySQL

WC: 113

To begin the process of converting my ERD into a functional database, an instance of MySQL was set up within a Docker container using the script shown below:

```yaml
services:
  db:
    image: mysql:9.2.0
    container_name: test-db
    ports:
      - "3306:3306"
    environment:
      - MYSQL_ROOT_PASSWORD=password
      - MYSQL_DATABASE=test-db
    volumes:
      - ./test-db:/var/lib/mysql
```

Figure 1: docker-compose.yml (Used to launch a Dockerized MySQL instance)

This approach also enabled me to keep the version of MySQL consistent, which is important going forward, as future version changes could introduce compatibility issues or even break functionality. It also allowed me to work across multiple machines without needing to install MySQL locally, as containerised environments ensure repeatable deployments and version control (Merkel, 2014). Where relevant, this report will discuss the differences that containerisation introduced in certain operations, and explain how the same tasks can also be performed on a MySQL instance running locally.

Creating the schema

WC: 364

The ERD produced in part one was largely accurate; however, during implementation, several changes and additions were made. For example, the message table was missing a field to store message content, which was added before proceeding. Additional attributes such as active_account and deleted_on were added to the account table to support soft deletion of user accounts for legal or data retention purposes, in line with GDPR requirements for the right to erasure (Information Commissioner's Office, 2021). MySQL Workbench was used to visualise the updated ERD and served as a reference when building the schema.

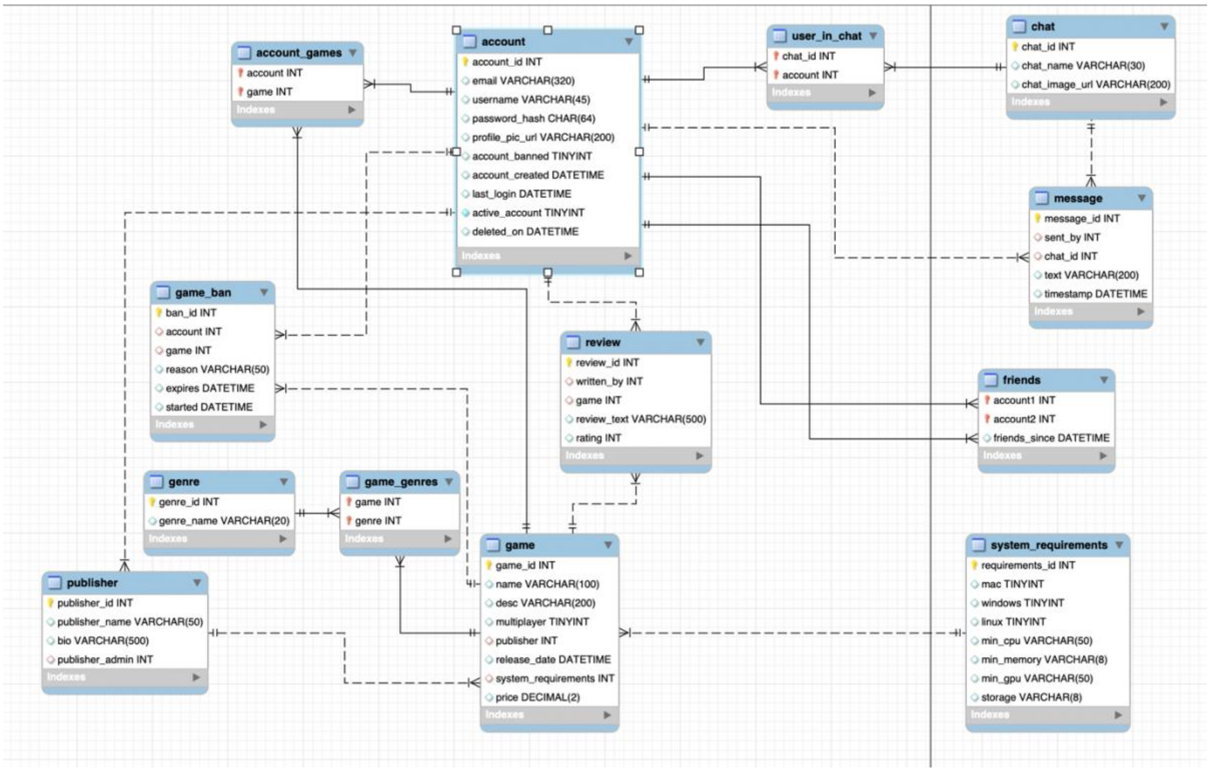

Figure 2: Finalised ERD for the Marketplace Database (From MySQL Workbench)

The schema was created incrementally, beginning with core entities such as account, followed by dependent tables like publisher, game, and review, which rely on foreign keys. To avoid constraint errors, parent tables were created before any child tables referencing them, as stipulated in the MySQL documentation (Oracle, 2024).

A public_account view was introduced to expose non-sensitive profile data while hiding private information such as email addresses and password hashes, making it useful for limiting the scope of data being retrieved. (Vanier et al., 2019, p. 234)

```
DROP DATABASE IF EXISTS marketplace;
CREATE DATABASE IF NOT EXISTS marketplace;
USE marketplace;
```

```
CREATE TABLE IF NOT EXISTS `account` (
  `account_id` INT NOT NULL AUTO_INCREMENT,
  `email` VARCHAR(320) NOT NULL,
  `username` VARCHAR(45) NOT NULL,
  `password_hash` CHAR(64) NOT NULL, -- SHA-256
  `profile_pic_url` VARCHAR(200) NULL,
  `account_banned` TINYINT NULL,
  `account_created` DATETIME NULL,
  `last_login` DATETIME NULL,
  `active_account` TINYINT NOT NULL DEFAULT 1,
  `deleted_on` DATETIME NULL,
  PRIMARY KEY (`account_id`),
  UNIQUE INDEX `email_UNIQUE` (`email` ASC) VISIBLE,
  UNIQUE INDEX `account_id_UNIQUE` (`account_id` ASC));
```

```
CREATE VIEW public_account AS
    SELECT `account_id`, `username`, `profile_pic_url`, `last_login`
    FROM `account`;
```

Figure 3: Snippet from final schema, showing how the database is created, along with the account table.

Indexes were applied to improve query performance, particularly on foreign key columns and fields requiring uniqueness, such as email, username, and publisher_id. Additional indexes were created on frequently queried columns like publisher_admin in the publisher table and written_by in the review table to enhance join efficiency (Vanier et al., 2019, pp. 48–49)

```sql
CREATE TABLE IF NOT EXISTS `publisher` (
  `publisher_id` INT NOT NULL AUTO_INCREMENT,
  `publisher_name` VARCHAR(50) NOT NULL,
  `bio` VARCHAR(500) NULL,
  `publisher_admin` INT NULL,
  PRIMARY KEY (`publisher_id`),
  UNIQUE INDEX `publisher_id_UNIQUE` (`publisher_id` ASC) VISIBLE,
  INDEX `publisher_admin_idx` (`publisher_admin` ASC) VISIBLE,
  CONSTRAINT `publisher_admin`
    FOREIGN KEY (`publisher_admin`)
    REFERENCES `marketplace`.`account` (`account_id`)
    ON DELETE SET NULL
    ON UPDATE CASCADE);
```

Figure 4: Publisher create script snippet, showing index on the publisher ID column / foreign key for more efficient joins.

Several stored procedures were developed to encapsulate common operations, including updating profile pictures, retrieving friends, and finding shared or unowned games. More complex logic was handled using Common Table Expressions (CTEs) with WITH … AS (…) syntax to simplify multi-step queries and improve readability. These procedures also made use of joins, subqueries, and the public_account view (Vanier et al., 2019, pp. 6, 233–234).

```sql
-- Stored Procedure to find common games
DROP PROCEDURE IF EXISTS FindCommonGames;
DELIMITER $$
CREATE PROCEDURE FindCommonGames(IN user1_param VARCHAR(45), IN user2_param VARCHAR(45))
BEGIN
    WITH user1_games AS (
        SELECT game
        FROM account_games
        WHERE account = (SELECT account_id FROM public_account WHERE username = user1_param)
    ),
    user2_games AS (
        SELECT game
        FROM account_games
        WHERE account = (SELECT account_id FROM public_account WHERE username = user2_param)
    ),
    shared_games AS (
        SELECT game
        FROM user1_games
        INNER JOIN user2_games USING (game)
    )
    SELECT g.*
    FROM game g
    JOIN shared_games sg ON g.game_id = sg.game;
END$$
DELIMITER ;
```

Figure 5: More complex stored procedure using WITH …AS syntax, used to find common games owned between two different players.

Role-based access control was implemented using MySQL GRANT statements. Three roles were defined: market_admin (full privileges, localhost-only access), app_user (read/write access), and readonly_user (read-only access). This allowed for more secure and maintainable permission management in a multi-user environment, aligning with MySQL 8's support for user-defined roles and privilege assignment. Referential integrity was enforced using foreign keys with appropriate cascading actions. For example, ON DELETE CASCADE was used for review and account_games, while ON DELETE SET NULL was applied to fields like publisher_admin, where retaining historical context was important (Vanier et al., 2019, pp. 233–236).

```sql
CREATE USER 'app_user'@'%' IDENTIFIED BY 'strongpassword';
GRANT SELECT, INSERT, UPDATE, DELETE ON marketplace.* TO 'app_user'@'%';

-- Read only user for analytics and read-only dashboards etc
CREATE USER 'readonly_user'@'%' IDENTIFIED BY 'readonlypass';
GRANT SELECT ON marketplace.* TO 'readonly_user'@'%';

-- Admin User, to used by developers that need full control
CREATE USER 'market_admin'@'127.0.0.1' IDENTIFIED BY 'supersecure';
GRANT ALL PRIVILEGES ON marketplace.* TO 'market_admin'@'127.0.0.1' WITH GRANT OPTION;
```

Figure 6: User roles within the create schema

Testing with data and querying

WC: 258

Realistic test data was created to support development and validate database functionality. An initial set of manually crafted records covered key scenarios, including users owning multiple games, publishers managing several titles, and reviews with varied ratings. This data included meaningful values such as usernames, profile pictures, descriptive game details, and system requirements.

```sql
-- Publishers
INSERT IGNORE INTO publisher (publisher_name, bio, publisher_admin) VALUES
( publisher_name 'Indie Galaxy',  bio 'We publish creative and story-rich indie games.',  publisher_admin 2),
( publisher_name 'NextGen Games',  bio 'AAA game studio pushing the boundaries of realism.',  publisher_admin 1),
( publisher_name 'Mojang',  bio 'We sold our souls to Microsoft!!!',  publisher_admin 3);

-- System Requirements
INSERT IGNORE INTO system_requirements (mac, windows, linux, min_cpu, min_memory, min_gpu, storage) VALUES
( mac 1,  windows 1,  linux 0,  min_cpu 'Intel i5',  min_memory '8GB',  min_gpu 'GTX 1050',  storage '20GB'),
( mac 0,  windows 1,  linux 1,  min_cpu 'AMD Ryzen 5',  min_memory '16GB',  min_gpu 'RTX 2060',  storage '50GB');

-- Games
INSERT IGNORE INTO game (name, `desc`, multiplayer, publisher, release_date, system_requirements, price) VALUES
( name 'Space Quest',  desc 'Explore galaxies and build civilizations.',  multiplayer 1,  publisher 1,  release_date '2023-10-01',  system_requirements 1,  price 29.99),
( name 'Zombie Dawn',  desc 'Survive in a post-apocalyptic world of the undead.',  multiplayer 0,  publisher 2,  release_date '2024-01-15',  system_requirements 2,  price 49.99),
( name 'Minecraft',  desc 'The cake is a lie',  multiplayer 1,  publisher 3,  release_date '2011-01-01',  system_requirements 1,  price 12.99);
```

Figure 7: Snippet of manually crafted test data for account, game, and review tables.

To scale the dataset, a Python script using the Faker library was used to generate over 100 accounts, 50 games, 10 publishers, and hundreds of related records, such as reviews, friendships, chat messages, and game ownership data. Data complexity was increased through special characters, varied data types, and enforced foreign key constraints to maintain referential integrity.

```python
# Accounts
f.write("-- Accounts\n")
for i in range(NUM_ACCOUNTS):
    email = fake.unique.email()
    username = fake.unique.user_name()
    password_hash = 'b06779c4959295605d26f9e7eb06356d5b5c6b1bd10cc28da75931f06997c601'  # static hash
    profile_pic_url = f"'{fake.image_url() if random.choice([True, False]) else 'NULL'}'"
    f.write(f"""INSERT IGNORE INTO account (email, username, password_hash, profile_pic_url, account_banned, account_created, last_login)
            VALUES ('{email}', '{username}', '{password_hash}', {profile_pic_url if profile_pic_url != 'NULL' else 'NULL'}, 0, NOW(), NOW());\n""")
```

Figure 8: Snippet from Python test data generation script using Faker to generate accounts.

A range of SQL queries were written to test functionality and performance, including SELECT statements with JOINs, subqueries, and Common Table Expressions (CTEs) for more complex logic. Aggregate functions such as COUNT() were used to generate meaningful results. Query performance was analysed using EXPLAIN, confirming efficient use of indexes on key fields like username, email, and foreign keys. As shown in Figure 9, the execution plan demonstrates index usage on the account_games and account tables, with MySQL using keys such as PRIMARY and username, significantly reducing the number of rows scanned and improving performance. (Vanier et al., 2019, pp. 42–44, 66–67)

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	PRIMARY	account_games	<null>	ref	PRIMARY,game_idx	PRIMARY	4	const	1	100	Using index
1	PRIMARY	account_games	<null>	eq_ref	PRIMARY,game_idx	PRIMARY	8	const,marketplace.account_games.game	1	100	Using index
1	PRIMARY	g	<null>	eq_ref	PRIMARY,game_id_UNIQUE	PRIMARY	4	marketplace.account_games.game	1	100	<null>
6	SUBQUERY	account	<null>	const	username	username	182	const	1	100	Using index
4	SUBQUERY	account	<null>	const	username	username	182	const	1	100	Using index

Figure 9: Screenshot of EXPLAIN plan showing use of indexes in query execution.

Data export and backup were automated using a Python script that ran mysqldump within the Docker container using docker exec, saving the output to a .sql file on the host machine. On a local installation of MySQL, the same process could be performed directly using:

`mysqldump -u root -p marketplace > marketplace_dump.sql`

and restored using:

`mysql -u root -p marketplace < marketplace_dump.sql`

ALTER TABLE statements were used during development to adapt the schema, and queries were incrementally tested to verify correctness and data integrity. Once confirmed, the changed where applied to the create script.

Challenges and Problem solving

WC: 127

Several challenges were encountered during development, particularly around maintaining referential integrity when inserting test data. Early attempts to insert records without considering foreign key dependencies led to constraint violations. This was resolved by ordering inserts carefully — creating parent records (e.g., account, publisher) before dependent tables like game or review.

Data type mismatches also caused errors, especially when inserting NULL values or exceeding defined VARCHAR lengths. These were resolved through validation and, where necessary, adjusting the schema.

Debugging was primarily carried out using JetBrains DataGrip, supported by incremental testing of queries. DataGrip provided an advanced query editor and better integration with my development environment compared to MySQL Workbench.

Additionally, the SHOW WARNINGS and SHOW GRANTS commands were particularly useful for diagnosing permission-related issues and resolving errors during testing.

Conclusion

WC: 123

Overall, a fully functional database that covers the original requirements set out in part one was developed. Decisions such as enforcing foreign keys, applying cascading behaviours appropriately, and using indexing on frequently queried fields helped data integrity and ensuring efficient query performance. The use of views and stored procedures provided a secure and maintainable way to encapsulate database logic and protect sensitive data.

Future improvements could include expanding security features, such as implementing password salting and additional input validation. As the database scales, further performance optimisations could be made my analysing index usage and optimising were appropriate. Additionally, the automated data generation process could also be extended to simulate more complex user behaviour and larger datasets to support load testing or analytics requirements.

References

- Beaulieu, A., 2009. *Learning SQL*. 2nd ed. Sebastopol, CA: O'Reilly Media, p.231.
- Elmasri, R. and Navathe, S.B., 2015. *Fundamentals of Database Systems*. 6th ed. Boston: Pearson, pp.633–639.
- Hamidi, A., Hamraz, A.R. and Rahmani, K., 2024. Database security mechanisms in MySQL. *Afghanistan Research Journal - Natural Science*, 4(1), pp.1–9. Available at: https://www.arj.af/index.php/arj/article/view/66 [Accessed 13 Apr. 2025].
- Information Commissioner's Office (ICO), 2021. *Right to erasure ('right to be forgotten')*. [online] Available at: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/individual-rights/individual-rights/right-to-erasure/ [Accessed 13 Apr. 2025].
- JetBrains, 2025. *DataGrip*. Available at: https://www.jetbrains.com/datagrip/ [Accessed 13 Apr. 2025].

- Merkel, D., 2014. Docker: Lightweight Linux containers for consistent development and deployment. *Linux Journal*, 2014(239), pp.2–2. Available at: https://www.linuxjournal.com/content/docker-lightweight-linux-containers-consistent-development-and-deployment [Accessed 13 Apr. 2025].
- Oracle, 2024. *MySQL 8.0 Reference Manual: 15.1.20.5 FOREIGN KEY Constraints*. [online] Available at: https://dev.mysql.com/doc/refman/8.0/en/create-table-foreign-keys.html [Accessed 13 Apr. 2025].
- Oracle, 2025. *MySQL Workbench*. Available at: https://www.mysql.com/products/workbench/ [Accessed 13 Apr. 2025].
- Vanier, E., Shah, B. and Malepati, T., 2019. *Advanced MySQL 8: Discover the Full Potential of MySQL and Ensure High Performance of Your Database*. Birmingham: Packt Publishing. Available at: http://ebookcentral.proquest.com/lib/ucsl/detail.action?docID=5675584 [Accessed 17 Apr. 2025].

Appendix

Appendix A: Output from running create script

	Time	Action	Response	Duration / Fetch Time
1	01:54:32	DROP DATABASE IF EXISTS marketplace	14 row(s) affected	0.046 sec
2	01:54:32	CREATE DATABASE IF NOT EXISTS marketplace	1 row(s) affected	0.0015 sec
3	01:54:32	USE marketplace	0 row(s) affected	0.00036 sec
4	01:54:32	DROP TABLE IF EXISTS `account`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.0010 sec
5	01:54:32	CREATE TABLE IF NOT EXISTS `account` (`account_id` INT NOT NUL...	0 row(s) affected	0.0077 sec
6	01:54:32	CREATE VIEW public_account AS SELECT `account_id`, `username`, `...	0 row(s) affected	0.0019 sec
7	01:54:32	DROP TABLE IF EXISTS `publisher`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.00080 sec
8	01:54:32	CREATE TABLE IF NOT EXISTS `publisher` (`publisher_id` INT NOT N...	0 row(s) affected	0.011 sec
9	01:54:32	DROP TABLE IF EXISTS `system_requirements`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.00089 sec
10	01:54:32	CREATE TABLE IF NOT EXISTS `system_requirements` (`requirement...	0 row(s) affected	0.0045 sec
11	01:54:32	DROP TABLE IF EXISTS `game`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.00062 sec
12	01:54:32	CREATE TABLE IF NOT EXISTS `game` (`game_id` INT NOT NULL AU...	0 row(s) affected	0.0098 sec
13	01:54:32	DROP TABLE IF EXISTS `review`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.00083 sec
14	01:54:32	CREATE TABLE IF NOT EXISTS `review` (`review_id` INT NOT NULL A...	0 row(s) affected	0.0086 sec
15	01:54:32	DROP TABLE IF EXISTS `genre`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.00077 sec
16	01:54:32	CREATE TABLE IF NOT EXISTS `genre` (`genre_id` INT NOT NULL AU...	0 row(s) affected	0.0078 sec
17	01:54:32	DROP TABLE IF EXISTS `game_genres`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.00068 sec
18	01:54:32	CREATE TABLE IF NOT EXISTS `game_genres` (`game` INT NOT NUL...	0 row(s) affected	0.0068 sec
19	01:54:32	DROP TABLE IF EXISTS `game_ban`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.00079 sec
20	01:54:32	CREATE TABLE IF NOT EXISTS `game_ban` (`ban_id` INT NOT NULL...	0 row(s) affected	0.0094 sec
21	01:54:32	DROP TABLE IF EXISTS `account_games`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.00076 sec
22	01:54:32	CREATE TABLE IF NOT EXISTS `account_games` (`account` INT NOT...	0 row(s) affected	0.012 sec
23	01:54:32	DROP TABLE IF EXISTS `chat`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.0023 sec
24	01:54:32	CREATE TABLE IF NOT EXISTS `chat` (`chat_id` INT NOT NULL AUT...	0 row(s) affected	0.0038 sec
25	01:54:32	DROP TABLE IF EXISTS `user_in_chat`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.0013 sec
26	01:54:32	CREATE TABLE IF NOT EXISTS `user_in_chat` (`chat_id` INT NOT NU...	0 row(s) affected	0.0057 sec
27	01:54:32	DROP TABLE IF EXISTS `message`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.0016 sec
28	01:54:32	CREATE TABLE IF NOT EXISTS `message` (`message_id` INT NOT N...	0 row(s) affected	0.0060 sec
29	01:54:32	DROP TABLE IF EXISTS `friends`	0 row(s) affected, 1 warning(s): 1051 Unknown table '...	0.00068 sec
30	01:54:32	CREATE TABLE IF NOT EXISTS `friends` (`account1` INT NOT NULL,...	0 row(s) affected	0.0052 sec
31	01:54:32	DROP USER IF EXISTS 'app_user'@'%'	0 row(s) affected	0.0023 sec
32	01:54:32	DROP USER IF EXISTS 'readonly_user'@'%'	0 row(s) affected	0.00082 sec
33	01:54:32	DROP USER IF EXISTS 'market_admin'@'127.0.0.1'	0 row(s) affected	0.00084 sec
34	01:54:32	CREATE USER 'app_user'@'%' IDENTIFIED BY 'strongpassword'	0 row(s) affected	0.0014 sec
35	01:54:32	GRANT SELECT, INSERT, UPDATE, DELETE ON marketplace.* TO 'app_u...	0 row(s) affected	0.00077 sec
36	01:54:32	CREATE USER 'readonly_user'@'%' IDENTIFIED BY 'readonlypass'	0 row(s) affected	0.0017 sec
37	01:54:32	GRANT SELECT ON marketplace.* TO 'readonly_user'@'%'	0 row(s) affected	0.00077 sec
38	01:54:32	CREATE USER 'market_admin'@'127.0.0.1' IDENTIFIED BY 'supersecure'	0 row(s) affected	0.0012 sec
39	01:54:32	GRANT ALL PRIVILEGES ON marketplace.* TO 'market_admin'@'127.0.0...	0 row(s) affected	0.00078 sec

Appendix B: Output from running basic test data insert

	Time	Action	Response
1	01:57:31	USE marketplace	0 row(s) affected
2	01:57:31	INSERT IGNORE INTO account (email, username, password_hash, profile_pic_url, account_banned, account_created, la...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...
3	01:57:31	INSERT IGNORE INTO publisher (publisher_name, bio, publisher_admin) VALUES ('Indie Galaxy', 'We publish creative an...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...
4	01:57:31	INSERT IGNORE INTO system_requirements (mac, windows, linux, min_cpu, min_memory, min_gpu, storage) VALUES (1...	2 row(s) affected Records: 2 Duplicates: 0 Warnings...
5	01:57:31	INSERT IGNORE INTO game (name, \`desc\`, multiplayer, publisher, release_date, system_requirements, price) VALUES (...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...
6	01:57:31	INSERT IGNORE INTO genre (genre_name) VALUES ('Adventure'), ('Survival'), ('Sci-Fi'), ('Horror'), ('Shooter'), ('Strateg...	16 row(s) affected Records: 16 Duplicates: 0 Warnin...
7	01:57:31	INSERT IGNORE INTO game_genres (game, genre) VALUES (1, 1), (1, 3), (2, 2), (2, 4)	4 row(s) affected Records: 4 Duplicates: 0 Warnings...
8	01:57:31	INSERT INTO review (written_by, game, review_text, rating) VALUES (1, 1, 'Absolutely loved exploring planets!', 9), (3, 2,...	2 row(s) affected Records: 2 Duplicates: 0 Warnings...
9	01:57:31	INSERT IGNORE INTO account_games (account, game) VALUES (1, 1), (1, 2), (3, 2)	3 row(s) affected Records: 3 Duplicates: 0 Warnings...
10	01:57:31	INSERT INTO game_ban (account, game, reason, expires, started) VALUES (3, 2, 'Toxic behavior in reviews', DATE_ADD...	1 row(s) affected
11	01:57:31	INSERT INTO chat (chat_name, chat_image_url) VALUES ('Space Fans', 'https://example.com/chat/space.jpg'), ('Zombi...	2 row(s) affected Records: 2 Duplicates: 0 Warnings...
12	01:57:31	INSERT IGNORE INTO user_in_chat (chat_id, account) VALUES (1, 1), (1, 3), (2, 3)	3 row(s) affected Records: 3 Duplicates: 0 Warnings...
13	01:57:31	INSERT INTO message (sent_by, chat_id, text, timestamp) VALUES (1, 1, 'Anyone found the hidden galaxy yet?', NOW())...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...
14	01:57:31	INSERT IGNORE INTO friends (account1, account2, friends_since) VALUES (1, 3, NOW())	1 row(s) affected

Appendix C: Output from running generated test data insert

	Time	Action	Response	Duration / Fe
1218	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (12, 10, 'Citizen possible able wall again...	1 row(s) affected	0.0013 sec
1219	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (25, 12, 'Shoulder plant husband church.'...	1 row(s) affected	0.0013 sec
1220	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (67, 15, 'Tonight hand note road certain...	1 row(s) affected	0.00072 sec
1221	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (80, 20, 'Hour finish increase interview e...	1 row(s) affected	0.00066 sec
1222	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (7, 11, 'Notice what she send throw meth...	1 row(s) affected	0.00069 sec
1223	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (73, 6, 'Year cold above check seven aga...	1 row(s) affected	0.0021 sec
1224	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (62, 16, 'Front responsibility need ten he...	1 row(s) affected	0.0015 sec
1225	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (40, 15, 'Or thing as avoid reflect teach.',...	1 row(s) affected	0.00071 sec
1226	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (87, 1, 'Hot member yet similar floor gen...	1 row(s) affected	0.0017 sec
1227	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (11, 15, 'Radio thank fine clear.', NOW())	1 row(s) affected	0.00067 sec
1228	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (38, 5, 'Million significant card ever assu...	1 row(s) affected	0.00069 sec
1229	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (32, 1, 'Out own certain party.', NOW())	1 row(s) affected	0.00063 sec
1230	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (43, 7, 'Far art magazine turn center her...	1 row(s) affected	0.00064 sec
1231	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (87, 8, 'Speak good that shoulder mome...	1 row(s) affected	0.0020 sec
1232	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (71, 7, 'Culture their local eye indeed car...	1 row(s) affected	0.00065 sec
1233	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (45, 3, 'Practice measure alone like disc...	1 row(s) affected	0.00064 sec
1234	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (97, 13, 'Among population allow.', NOW())	1 row(s) affected	0.00064 sec
1235	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (84, 1, 'Assume record discuss stay.', NO...	1 row(s) affected	0.00061 sec
1236	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (83, 7, 'Send black break area traditional...	1 row(s) affected	0.00060 sec
1237	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (39, 15, 'Speak head report out ready ra...	1 row(s) affected	0.00048 sec
1238	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (80, 1, 'Soon can election upon manage...	1 row(s) affected	0.00060 sec
1239	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (81, 2, 'Development box including busin...	1 row(s) affected	0.00073 sec
1240	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (43, 1, 'Player argue knowledge drop.', N...	1 row(s) affected	0.0015 sec
1241	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (22, 16, 'Five big cause administration n...	1 row(s) affected	0.00069 sec
1242	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (17, 13, 'Whom stay government.', NOW())	1 row(s) affected	0.00099 sec
1243	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (33, 14, 'Finish ahead those purpose.', N...	1 row(s) affected	0.00061 sec
1244	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (15, 18, 'Learn north key total.', NOW())	1 row(s) affected	0.00047 sec
1245	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (78, 6, 'Yeah election serve several avail...	1 row(s) affected	0.0017 sec
1246	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (93, 7, 'Able nothing usually dog movie h...	1 row(s) affected	0.00066 sec
1247	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (63, 3, 'Audience local possible employe...	1 row(s) affected	0.00062 sec
1248	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (2, 11, 'New serve risk research least real...	1 row(s) affected	0.00064 sec
1249	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (67, 5, 'Anything performance in hour ind...	1 row(s) affected	0.00065 sec
1250	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (9, 17, 'Young half address think.', NOW())	1 row(s) affected	0.00066 sec
1251	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (48, 17, 'But phone he arm blood make.',...	1 row(s) affected	0.0020 sec
1252	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (20, 1, 'Leg relationship order.', NOW())	1 row(s) affected	0.00061 sec
1253	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (9, 6, 'Clear move better.', NOW())	1 row(s) affected	0.00046 sec
1254	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (81, 6, 'Director tend certainly you admin...	1 row(s) affected	0.00063 sec
1255	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (9, 6, 'Cover kitchen interesting early wit...	1 row(s) affected	0.00065 sec
1256	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (49, 2, 'Blood challenge history person...	1 row(s) affected	0.00062 sec
1257	01:58:56	INSERT IGNORE INTO message (sent_by, chat_id, text, timestamp) VALUES (11, 9, 'Turn reveal son white claim how.',...	1 row(s) affected	0.00063 sec

Appendix D: Output from running create stored procedures

	Time	Action	Response	Duration / Fe
1	01:59:57	USE marketplace	0 row(s) affected	0.0013 sec
2	01:59:57	DROP PROCEDURE IF EXISTS UpdateAccountPicture	0 row(s) affected, 1 warning(s): 1305 PROCEDURE m...	0.0027 sec
3	01:59:57	CREATE PROCEDURE UpdateAccountPicture(IN username_param VARCHAR(45), IN hpassword_param CH...	0 row(s) affected	0.0034 sec
4	01:59:57	-- Stored Procedure to find common games DROP PROCEDURE IF EXISTS FindCommonGames	0 row(s) affected, 1 warning(s): 1305 PROCEDURE m...	0.0017 sec
5	01:59:57	CREATE PROCEDURE FindCommonGames(IN user1_param VARCHAR(45), IN user2_param VARCHAR(45))...	0 row(s) affected	0.0036 sec
6	01:59:57	-- Stored procedure to find unowned games DROP PROCEDURE IF EXISTS FindUnownedGamesByID	0 row(s) affected, 1 warning(s): 1305 PROCEDURE m...	0.0019 sec
7	01:59:57	CREATE PROCEDURE FindUnownedGamesByID(IN account_id_param INT) BEGIN WITH games_owned AS...	0 row(s) affected	0.0021 sec
8	01:59:57	-- Returns a list of friends an account has DROP PROCEDURE IF EXISTS GetFriendsByAccountID	0 row(s) affected, 1 warning(s): 1305 PROCEDURE m...	0.0015 sec
9	01:59:57	CREATE PROCEDURE GetFriendsByAccountID(IN account_id_param INT) BEGIN -- Find friend IDs WIT...	0 row(s) affected	0.0021 sec
10	01:59:57	-- Soft delete account in accordance with UK GDPR DROP PROCEDURE IF EXISTS SoftAccountDelete	0 row(s) affected, 1 warning(s): 1305 PROCEDURE m...	0.0023 sec
11	01:59:57	CREATE PROCEDURE SoftAccountDelete(IN account_id_param INT, IN hashed_email TEXT) BEGIN UPDA...	0 row(s) affected	0.0042 sec
12	01:59:57	-- Full Delete after data retention period (1-2 years for audit purposes) DROP PROCEDURE IF EXISTS FullA...	0 row(s) affected, 1 warning(s): 1305 PROCEDURE m...	0.0015 sec
13	01:59:57	CREATE PROCEDURE FullAccountDelete(IN account_id_param INT) BEGIN DELETE FROM account WHER...	0 row(s) affected	0.0021 sec

Appendix E: Console output showing successful run of select queries

	Time	Action	Response	Duration / Fetch Time
1	02:02:03	USE marketplace	0 row(s) affected	0.00089 sec
2	02:02:03	SHOW GRANTS FOR 'root'@'localhost'	3 row(s) returned	0.00085 sec / 0.000...
3	02:02:03	SHOW GRANTS FOR 'app_user'@'%'	2 row(s) returned	0.00085 sec / 0.000...
4	02:02:03	SHOW GRANTS FOR 'readonly_user'@'%'	2 row(s) returned	0.00073 sec / 0.0000...
5	02:02:03	SHOW GRANTS FOR 'market_admin'@'127.0.0.1'	2 row(s) returned	0.00047 sec / 0.0000...
6	02:02:03	SELECT count(account_id) from public_account LIMIT 0, 1000	1 row(s) returned	0.0012 sec / 0.00000...
7	02:02:03	SELECT * FROM public_account LIMIT 0, 1000	103 row(s) returned	0.0011 sec / 0.00001...
8	02:02:03	SELECT * FROM public_account WHERE username = 'AliceGamer' LIMIT 0, 1000	1 row(s) returned	0.00091 sec / 0.0000...
9	02:02:03	SELECT genre_name FROM genre LIMIT 0, 1000	32 row(s) returned	0.00048 sec / 0.000...
10	02:02:03	SELECT * FROM game LIMIT 0, 1000	53 row(s) returned	0.0012 sec / 0.00001...
11	02:02:03	WITH user_id AS (SELECT account_id FROM account WHERE username = 'AliceGamer'), games_ids_owne...	3 row(s) returned	0.0011 sec / 0.00000...
12	02:02:03	WITH games_ids_owned AS (SELECT game FROM account_games WHERE account IN (SELECT account_id...	3 row(s) returned	0.0013 sec / 0.00000...
13	02:02:03	WITH games_ids_owned AS (SELECT game FROM account_games WHERE account IN (SELECT account_id...	1 row(s) returned	0.0010 sec / 0.00000...
14	02:02:03	CALL FindCommonGames('AliceGamer', 'CarolPlays')	1 row(s) returned	0.00070 sec / 0.0000...
15	02:02:03	EXPLAIN WITH user1_games AS (SELECT game FROM account_games WHERE account = (SELECT accou...	5 row(s) returned	0.0014 sec / 0.00000...
16	02:02:03	CALL FindUnownedGamesByID(1)	50 row(s) returned	0.0014 sec / 0.00001...
17	02:02:03	CALL FindUnownedGamesByID(2)	50 row(s) returned	0.0013 sec / 0.00000...
18	02:02:03	CALL FindUnownedGamesByID(3)	52 row(s) returned	0.00086 sec / 0.000...
19	02:02:03	SELECT * FROM public_account WHERE account_id = (SELECT account_id FROM account WHERE us...	1 row(s) returned	0.0013 sec / 0.00000...
20	02:02:03	CALL GetFriendsByAccountID(1)	4 row(s) returned	0.00060 sec / 0.000...
21	02:02:03	CALL GetFriendsByAccountID(2)	4 row(s) returned	0.00093 sec / 0.000...
22	02:02:03	CALL GetFriendsByAccountID(3)	2 row(s) returned	0.00079 sec / 0.0000...

Appendix F: Result of stored procedure call

```
CALL GetFriendsByAccountID(1);
```

account_id	email	username	password_hash	profile_pic_url	account_bann...	account_created	last_login	active_accou...
3	carol@example.com	CarolPlays	b06779c4959295605d26f9e7eb06356d5b5c6b1...	NULL	0	2025-04-21 00:57:31	2025-04-21 00:57:31	1
12	langangela@example.net	pkelly	b06779c4959295605d26f9e7eb06356d5b5c6b1...	https://picsum.photos/948/1002	0	2025-04-21 00:58:54	2025-04-21 00:58:54	1
21	wbuchanan@example.com	reginahartman	b06779c4959295605d26f9e7eb06356d5b5c6b1...	https://dummyimage.com/408x257	0	2025-04-21 00:58:54	2025-04-21 00:58:54	1
62	joshua04@example.org	harellano	b06779c4959295605d26f9e7eb06356d5b5c6b1...	NULL	0	2025-04-21 00:58:54	2025-04-21 00:58:54	1

Appendix G: Output from running update queries

	Time	Action	Response	Duration / Fetch Time
1	02:03:39	USE marketplace	0 row(s) affected	0.0012 sec
2	02:03:39	UPDATE account SET email = 'aliceupdated@example.com' WHERE username = 'AliceGamer' AND password_has...	1 row(s) affected Rows matched: 1 Changed: 1 Warni...	0.0053 sec
3	02:03:39	CALL UpdateAccountPicture('AliceGamer', 'b06779c4959295605d26f9e7eb06356d5b5c6b1bd10cc28da75931f...	1 row(s) affected	0.0030 sec
4	02:03:39	UPDATE account SET last_login = NOW() WHERE username = 'BobDev' AND password_hash = 'b06779c49592...	1 row(s) affected Rows matched: 1 Changed: 1 Warni...	0.0027 sec

Appendix H: Output from running delete queries

	Time	Action	Response	Duration / Fetch Time
1	02:04:44	USE marketplace	0 row(s) affected	0.0020 sec
2	02:04:44	DELETE FROM friends WHERE (account1 = LEAST(1, 3) AND account2 = GREATEST(1, 3))	1 row(s) affected	0.0042 sec
3	02:04:44	CALL SoftAccountDelete(2, 'example_email_hash')	1 row(s) affected	0.0026 sec
4	02:04:44	CALL FullAccountDelete(2)	1 row(s) affected	0.0023 sec